

APEX ENGINEERING INSTITUTE

Department of Mechanical & Civil Engineering

ENGR 230: Engineering Statics & Dynamics**Problem Set 4: Rigid Body Equilibrium & Particle Kinematics** **Date Issued:** Oct 7, 2026 **Due Date:** Oct 14, 2026 **Instructor:** Prof. Marcus Vance **Total Points:** 100 Points

Student	Name:	Student	ID:	Section	/	Recitation:
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⚠ Submission Guidelines & Policy: Show all free-body diagrams (FBDs), coordinate axis definitions, vector components, and intermediate algebraic steps clearly. State all physical assumptions (e.g., rigid members, frictionless pins, uniform gravitational field with $g = 9.81 \text{ m/s}^2$). Express final numerical answers with appropriate physical units and 3 significant figures.